Ex: $A = 2$

$$\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\Rightarrow -2x_1 - 2x_3 = 0$

$x_1 + x_3 = 0$

$x_1 + x_3 = 0$

$\Rightarrow x_1 + x_3 = 0$

~~$x_1 + x_3 = 0$~~

$\Rightarrow x_1 = -x_3$

Choose $x_2 = 0$

$$X = \begin{pmatrix} -x_3 \\ 0 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} x_3$$

Let us choose $x_3 = 1$

$$\therefore X = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Again choose $x_2 = 1$

$$X = \begin{pmatrix} -x_3 \\ 1 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} x_3$$

Choose $x_3 = 0$

$$\therefore X = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

P Kalika Maths